

The problem:

Capabilities beyond

what's built in

Introduction

From built-in tools to ecosystem tools

Your journey:

Part 1

Understood tools fundamentally and built your first tool-enabled agent

Part 2

Used production-ready built-in tools provided by ADK

Part 3

Connect to external tool servers via a standardized protocol

The shift:

Instead of using tools built into ADK, you connect to pre-existing tool servers maintained by the community and ecosystem.



The problem:

Capabilities beyond what's built in

In module 2, you learned that built-in tools save time versus writing custom integrations. But what about capabilities that:

Python

```
# You need capabilities like:  
# - File system access  
# - Database connections  
# - Third-party API integrations (Slack, Notion, GitHub)  
# - Specialized tools for specific domains  
  
# These aren't built into ADK!  
# Do you have to write everything yourself?
```

Issues with building everything from scratch:

- ✗ **Not built into ADK:** Many useful capabilities aren't included as built-in tools
- ✗ **Already exist as open-source tools:** Someone else has already solved these problems
- ✗ **Require complex implementation:** Real-world integrations need protocols, auth, error handling
- ✗ **Maintained by experts:** Tool-specific knowledge you don't need to learn

The root problem

There's a huge ecosystem of pre-built tools, but how do you connect them to your ADK agent?

